

Image Generation Models

Project: Verified Study Buddy

1. Your mission

Your learning target

I can explain the mechanism and terminology of image generation and provenance.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Prompt Evaluation. Give an example and a mistake that the idea helps you avoid.

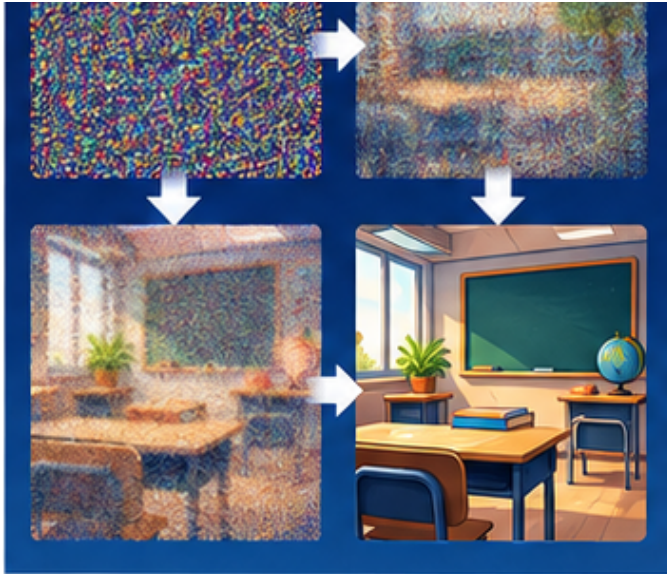
Without opening the old lesson, explain one key idea from Iterative Prompting. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Sampling and Variability. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 10: Image Generation Models

Image generation iteratively constructs pixels guided by learned representations.



CREATOR MISSION

Sequence a simplified noise-to-image card model.

Creator name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

Recall Generation failures and verification. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

This week's end goal

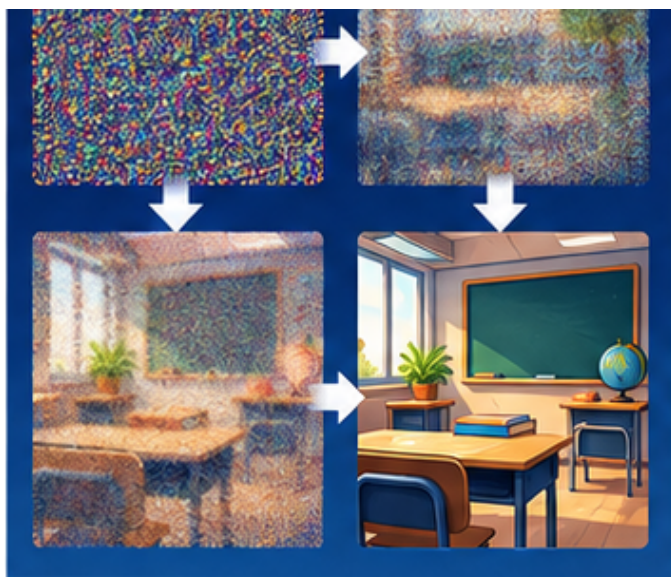
By the end, I can describe a toy denoising process and distinguish generated illustrations from observations.

2. Notice the evidence

PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use image generation models to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

An image generator constructs an output from learned patterns and a specification. A generated picture is not a record that the depicted event occurred.

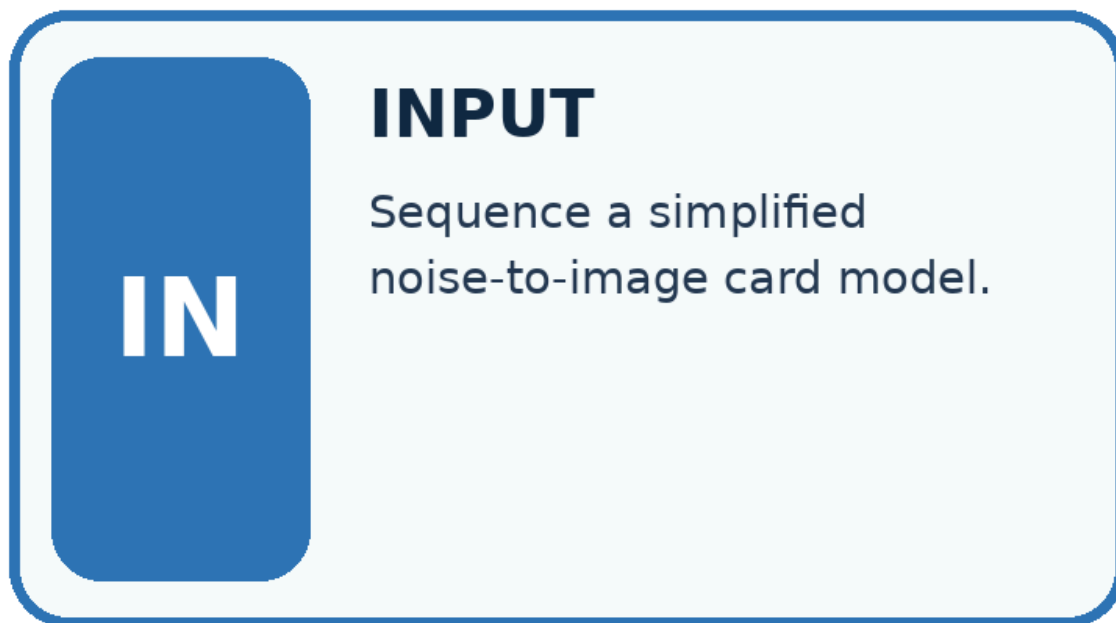
Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓



SYSTEM

Image generation iteratively constructs pixels guided by learned representations.

↓



EVIDENCE

Sequence a simplified noise-to-image card model.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Keep a provenance record containing specification, tool/version when known, generation date, edits, permissions and intended use. Label fictional imagery. Inspect visual requirements and avoid inferring identity or event authenticity from appearance.

Image generation and provenance — evidence trace

Input and scope · Check or calculation · Conclusion and limit

Input and scope: A fictional poster requires exactly three labelled circles. Generated image shows four circles and one unreadable label. A caption calls it a photograph of a real lesson.

Check or calculation: Keep a provenance record containing specification, tool/version when known, generation date, edits, permissions and intended use.

Conclusion and limit: It fails count and label criteria; the photograph claim is unsupported. Record it as a generated fictional illustration and revise or reject before use.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

CASE GALLERY

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.



NORMAL
Expected learner request



AMBIGUOUS
Two plausible meanings

NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



MISSING SOURCE

Evidence unavailable



ADVERSARIAL

Rules or trust challenged

MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change: Relabel the illustration as a photograph of an actual laboratory measurement.

Prediction: That caption is unsupported.

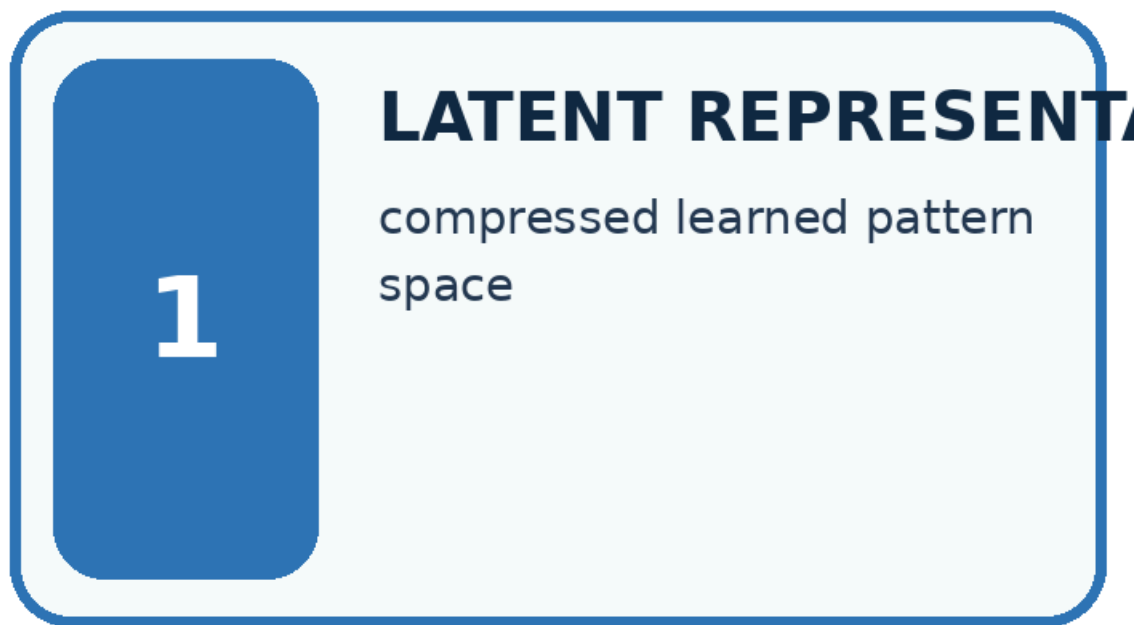
Reason: Its provenance is an invented diagram of a process, not a camera record of an event.

5. Words that unlock the idea

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



2

NOISE

random starting signal

3

DENOISING

iterative prediction toward
structure

4

CONDITIONING

guidance from prompt or reference

Word · My own example · Not this / boundary

Word: latent representation

My own example: An internal representation learned or constructed by a model rather than directly supplied as the original observation.

Not this / boundary: _____

Word: noise

My own example: Variation or interference that obscures the information relevant to a task.

Not this / boundary: _____

Word: denoising

My own example: Reducing noise in a representation; some generative models learn repeated denoising steps to create outputs.

Not this / boundary: _____

Word: conditioning

My own example: Making a model's output depend on supplied information, such as a prompt or image.

Not this / boundary: _____

Words used in the closer explanation

Word · Meaning

Word: Provenance

Meaning: Recorded origin and transformation history

Word: Visual criterion

Meaning: A stated requirement checked against an image

Word: Authenticity claim

Meaning: A claim about the source or reality of depicted content

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

Worked evidence for Image Generation Models

Step · Evidence and reasoning

Step: 1

Evidence and reasoning: In this simplified diffusion-style image generator, begin with random visual noise. Other image-generation methods use different procedures.

Step: 2

Evidence and reasoning: At each step, predict a slightly more structured image guided by the prompt.

Step: 3

Evidence and reasoning: Learned patterns influence shapes, textures, and relationships.

Step: 4

Evidence and reasoning: The result is generated, not retrieved evidence of a real event.

Trace these corrected steps. Keep the stated metric, assumptions and interpretation when you change the case.

1. In this simplified diffusion-style image generator, begin with random visual noise. Other image-generation methods use different procedures.
2. At each step, predict a slightly more structured image guided by the prompt.
3. Learned patterns influence shapes, textures, and relationships.
4. The result is generated, not retrieved evidence of a real event.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

A fictional poster requires exactly three labelled circles. Generated image shows four circles and one unreadable label. A caption calls it a photograph of a real lesson.

It fails count and label criteria; the photograph claim is unsupported. Record it as a generated fictional illustration and revise or reject before use.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Image Generation Models

A fictional lesson poster illustrates one family of image-generation approaches.

The facts we will use

Toy 2x2 noisy numerical grid: $[[0.2,0.8],[0.6,0.3]]$. A specified illustrative denoising step gives $[[0.1,0.9],[0.4,0.2]]$.

These supplied arrays illustrate steps only; they are not outputs from a real trained diffusion model.

AI ACADEMY / AI-3/10

Illustrative grids, not executed model outputs

	Left	Right
Before top	0.2	0.8
Before bottom	0.6	0.3
After top	0.1	0.9
After bottom	0.4	0.2

Read a cell using both its row and column labels.

Image Generation Models

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Noisy representation

Four toy numerical cells form the starting grid.



2. Condition on request

A text request guides the imagined learned denoising process.



3. Illustrative update

Use the supplied second grid to show one change in representation.



4. Check provenance

Caption it as a generated-process illustration, not a photographed experiment.

CHANGE ONE THING

Relabel the illustration as a photograph of an actual laboratory measurement.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Noisy representation

What happens and why: Four toy numerical cells form the starting grid.

Step: 2. Condition on request

What happens and why: A text request guides the imagined learned denoising process.

Step: 3. Illustrative update

What happens and why: Use the supplied second grid to show one change in representation.

Step: 4. Check provenance

What happens and why: Caption it as a generated-process illustration, not a photographed experiment.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Diffusion-style generators learn to remove noise in stages, often guided by text, but a real implementation can operate in a latent representation rather than direct pixels. The toy values let us inspect an update without pretending to simulate that trained process. An attractive final image still needs checks for labels, requested relations and factual claims.

What this example does not establish

Not all image generators use diffusion; this toy step is neither a complete algorithm nor an executed model.

Read the labelled picture: What may these two grids explain, and what can they not establish about a real generator?

Change one thing: Relabel the illustration as a photograph of an actual laboratory measurement. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



COMPARE • DIAGNOSE

Weak idea: An image model searches a hidden folder and copies the closest picture. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: works

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: stated rule

Actual: _____

Likely cause / next test: _____

Evidence record: missing input

Expected: safe response

Actual: _____

Likely cause / next test: _____

Evidence record: fresh transfer

Expected: generalizes

Actual: _____

Likely cause / next test: _____

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer. For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

8. Practise together

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SPECIFY THE SYSTEM

Define task, trusted context, constraints, schema, tools, checks, fallback, and stopping rule before trusting output.

SET task, audience, constraints, output_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF evidence missing: ask, retrieve, or abstain RETURN useful output + source trail + limitations

Trace one input. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Provenance tracks production history; it does not prove the content is accurate or establish every legal permission.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself. The first may require a correction; the second requires an explicitly revised method and new evidence. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Image generation and provenance”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Image generation and provenance. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

What may these two grids explain, and what can they not establish about a real generator?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



PREDICT → RUN → VERIFY → EXPLAIN

Order noise-to-image cards, vary one prompt condition, compare composition changes, and label what cannot be inferred as fact.

Order noise-to-image cards, vary one prompt condition, compare composition changes, and label what cannot be inferred as fact.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

Test / input: normal

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



CREATOR MASTERY
1 Define • 2 Trace • 3 Build • 4 Evaluate • 5 Verify • 6 Defend limits

1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or evidence path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

11. Connect your project

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



VERIFIED STUDY BUDDY

Treat generated diagrams as illustrations; pair factual claims with approved sources.

THIS WEEK’S CHECKPOINT Treat generated diagrams as illustrations; pair factual claims with approved sources.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → human decision

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under human control?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent output with verified truth.

Evidence for your project

Use Study Buddy generated diagrams only with clear provenance and checked educational labels.

12. Show what you know

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



INDEPENDENT CREATOR PROOF

Sequence a simplified noise-to-image card model.

CHALLENGE Sequence a simplified noise-to-image card model.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My evidence that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

13. Study a complete case

Week 10 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A generator creates a realistic fictional bicycle beside a tree.

The question

How does this differ from event photography?

Worked explanation

It synthesises an image from learned patterns and instructions; it does not document that event. Many systems denoise iteratively, but architectures vary.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

A generated image does not document an event

Evidence or stage · Value or result

Evidence or stage: Brief

Value or result: Bicycle beside a tree

Evidence or stage: Model

Value or result: Synthesises visual patterns

Evidence or stage: Image

Value or result: Fictional scene

Evidence or stage: Check

Value or result: Content and intended use

A generator creates a realistic fictional bicycle beside a tree.

How does this differ from event photography?

It synthesises an image from learned patterns and instructions; it does not document that event. Many systems denoise iteratively, but architectures vary.

14. Try a new case independently

Independent case: Apply a new case independently

A generated diagram requires two arrows and labels input/output. It has two arrows but swapped labels. Metadata records generation, yet the caption says experimental observation. Audit.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 10 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

An image model draws a plant with seven roots after a request for exactly four. What does this show, and where should an exact count diagram be made?

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Treat generated diagrams as illustrations; pair factual claims with approved sources.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

An image generator produces a convincing picture of a school greenhouse on Mars. A student calls it photographic evidence. Explain what the image can and cannot establish.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: A fictional lesson poster illustrates one family of image-generation approaches.

By the end, I can · How I will show it

By the end, I can: Explain the mechanism and terminology of image generation and provenance

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Apply the stated procedure to a new case with a traceable result

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Identify an assumption or limitation and design a revealing follow-up

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Explain the mechanism and terminology of image generation and provenance

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.

17. Supported practice, repair and portfolio

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional lesson poster illustrates one family of image-generation approaches.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Caption it as a generated-process illustration, not a photographed experiment.

1. Noisy representation
2. Condition on request
3. Check provenance
4. Illustrative update

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Check provenance Caption it as a generated-process illustration, not a photographed experiment. Diffusion-style generators learn to remove noise in stages, often guided by text, but a real implementation can operate in a latent representation rather than direct pixels. The toy values let us inspect an update without pretending to simulate that trained process. An attractive final image still needs checks for labels, requested relations and factual claims.

Practice 2: explain the deciding evidence

Changed-case practice: Relabel the illustration as a photograph of an actual laboratory measurement. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. That caption is unsupported.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: That caption is unsupported. Its provenance is an invented diagram of a process, not a camera record of an event.

Repair a missing building block

Review Prompt Evaluation (ai-3/9). Goal: Explain the mechanism and terminology of controlled prompt experiments. Short practice: Why is B's selection defensible here without calling it universally superior?. Return to this week and explain what you changed.

Review Iterative Prompting (ai-3/7). Goal: Explain the mechanism and terminology of controlled prompt experiments. Short practice: What evidence supports improvement here, and what remains unproven?. Return to this week and explain what you changed.

Review Sampling and Variability (ai-3/4). Goal: Sample from a distribution using explicit cumulative intervals. Short practice: Does $T=0.5$ make B impossible, and does choosing A guarantee factual accuracy?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.